

LEE CHONG WEI

Career Performance Analysis

Detailed Project Documentation

Excel • MySQL • SQL • Power BI • DAX

Table of Contents

TOPIC
• Project Overview
• 2. Project Objectives
• 3. Dataset & Data Preparation
• 4. Excel Analysis
• 5. MySQL Database
• 6. SQL Analysis
• 7. Power BI & DAX
• 8. Validation & Quality Checks
• 9. Key Findings
• 10. Project Outcome & Conclusion

1. Project Overview

This project analyzes the professional badminton match history of Lee Chong Wei using match-level data. The objective was to transform raw match records into meaningful insights about match results, opponents, tournaments, tournament tiers, rounds, and performance over time.

The project followed an end-to-end data analytics workflow: Excel, MySQL/SQL, and Power BI. The final validated dataset contains 415 matches, including 373 wins and 42 losses, resulting in an overall win rate of approximately 89.88% across 94 tournaments.

2. Project Objectives

- Analyze overall match performance.
- Calculate total matches, wins, losses, and win percentage.
- Analyze performance across different years and months.
- Identify frequently faced opponents and compare head-to-head performance.
- Analyze tournament participation and performance.
- Compare performance across tournament tiers.
- Analyze performance across tournament rounds.
- Examine finals performance.
- Build an interactive Power BI dashboard to present the analysis.

3. Dataset & Data Preparation

The original dataset contained badminton match-level records. Lee Chong Wei's men's singles records were isolated and transformed into a dedicated analytical dataset.

Column	Purpose
Match_Id	Unique match identifier
Date	Match date
Year / Month / Day	Date-derived fields
Tournament	Tournament name
Tier	Tournament category
Round	Tournament stage
Host_Location	Host/location information
Player	Lee Chong Wei
Opponent	Opposing player
Winner	Match winner
Result	Lee's Win/Loss result
Set_1_Score / Set_2_Score / Set_3_Score	Set-level scores

Data-quality checks included duplicate Match IDs, missing values, result consistency, round validation, and tier validation.

Validation	Result
Duplicate Match IDs	0
Missing Host_Location	30
Missing Set_2_Score	326
Missing Set_3_Score	326
Total Matches	415
Wins	373
Losses	42
Tournaments	94

Missing source information was retained as unavailable rather than being replaced with invented values. The master metrics were cross-validated in Excel, SQL, and Power BI.

4. Excel Analysis

Excel was the first analytical and validation stage of the project. It was used to inspect the cleaned data, perform calculations, explore patterns, and independently validate the results later reproduced in SQL and Power BI.

The analysis covered wins and losses, overall win percentage, year-wise and month-wise performance, tournament participation, tournament tiers, tournament rounds, opponent performance, and finals performance.

Overall win percentage was calculated as $\text{Wins} \div \text{Total Matches} \times 100$, producing approximately 89.88%. Excel was also used to confirm the master values of 415 matches, 373 wins, 42 losses, and 94 tournaments.

5. MySQL Database

After preparation in Excel, the cleaned dataset was imported into MySQL for structured database analysis. The final table was named **lee_singles**.

The table contained Match_Id, Date, Year, Month, Day, Tournament, Tier, Round, Host_Location, Player, Opponent, Winner, Result, Set_1_Score, Set_2_Score, and Set_3_Score.

The dataset was imported using the MySQL command-line approach and **LOAD DATA LOCAL INFILE**. The table structure and import settings were corrected and the final dataset was validated.

Validation included total record count, duplicate Match IDs, valid Win/Loss results, and checks for tournament, tier, and round values. The validated table then became the foundation for SQL analysis.

6. SQL Analysis

SQL was used to perform structured analysis of the match history and to progressively practice both fundamental and advanced analytical techniques.

Basic aggregation used COUNT(), SUM(), GROUP BY, and ORDER BY to calculate match counts, wins, losses, and grouped statistics.

Conditional analysis used CASE WHEN. For example:

```
Sum(CASE WHEN RESULT = "WIN" THEN 1 ELSE 0 END)
```

“HAVING” was used to filter aggregated groups after GROUP BY. This was particularly useful when requiring tournaments to have at least five matches.

CTEs (Common Table Expressions) were introduced to break complex analysis into logical stages. A typical structure was:

```
WITH tournament_stats AS (  
  SELECT Tournament, COUNT(*) AS TOTAL_MATCHES,  
  SUM(CASE WHEN Result = 'Win' THEN 1 ELSE 0 END) AS TOTAL_WINS  
  FROM lee_singles  
  GROUP BY Tournament  
)  
SELECT * FROM tournament_stats;
```

Window functions were then practiced using PARTITION BY and ORDER BY, including COUNT() OVER(), ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), and LEAD().

Concept	Purpose
GROUP BY	Aggregate by year, tournament, tier, round, or opponent
CASE WHEN	Conditional counting and classification
HAVING	Filter aggregated groups
CTEs	Break complex queries into logical steps
ROW_NUMBER()	Assign unique sequential positions
RANK()	Rank with ties and gaps
DENSE_RANK()	Rank with ties without gaps
LAG()	Compare with a previous row
LEAD()	Compare with a following row
PARTITION BY	Perform window calculations within groups

SQL analysis covered overall performance, year/month trends, tournament and tier performance, rounds, finals, head-to-head analysis, and ranking-based comparisons.

7. Power BI & DAX

Power BI was used to convert the validated SQL analysis into a three-page interactive dashboard. The MySQL table was connected through the MySQL connector and used as the report's data source.

The primary DAX measure was:

```
Win % Rate = DIVIDE( CALCULATE( COUNTROWS('lee lee_singles'), 'lee lee_singles'[Result] = "Win" ),
COUNTROWS('lee lee_singles'), 0 )
```

COUNTROWS() counts records; **CALCULATE()** applies the Win filter; and **DIVIDE()** calculates the percentage safely, returning 0 when the denominator is zero.

The same measure was reused under different filter contexts for overall, opponent, tournament, and finals win rates.

Page 1 — Career Overview: KPI cards for Total Matches, Wins, Losses, Win Rate and tournament information, plus year and tier analysis.

Page 2 — Opponent Analysis: bar charts for win/loss and win percentage by opponent, a Top 10 opponents analysis, a detailed match table, and opponent filtering.

Page 3 — Tournament & Round Analysis: tournament frequency, Win % by Tournament, Performance by Round, four finals KPI cards, and Year/Tournament/Tier/Round slicers.

The Win % by Tournament visual used a minimum five-match threshold so very small samples did not dominate comparisons.

Visual	Purpose
Card / KPI	Display a single important metric
Bar Chart	Compare categorical values
Column Chart	Compare ordered stages/categories
Table	Show detailed match records
Slicer	Provide interactive filtering

Dashboard theme: Background #0F172A; Cards #1E293B.

The report was published to the Power BI workspace **project_1**. The published report and semantic model were checked after publishing.

8. Validation & Quality Checks

Validation was performed across Excel, MySQL/SQL, Power BI, and the published Power BI environment.

Metric	Validated Value
Total Matches	415
Wins	373
Losses	42
Overall Win Rate	89.88%
Tournaments	94

Match_Id was checked for duplicates and returned 0 duplicate IDs. Result values were validated as Win/Loss. Missing values were identified and preserved as unavailable source information.

Power BI DAX results were checked against independently validated Excel and SQL results. Slicers and visual interactions were tested. The published report and semantic model in project_1 were also checked.

The tournament win-rate comparison used a five-match minimum. Genuine 100% win rates were retained where the underlying records supported them; they were not artificially removed or changed.

Finals validation produced 67 finals, 48 wins, 19 losses, and approximately 71.64% win rate. The consistency checks $48 + 19 = 67$ and $48 \div 67 \approx 71.64\%$ were confirmed.

9. Key Findings

The project contains 415 recorded matches, with 373 wins and 42 losses, giving an overall win rate of approximately 89.88%. The dataset covers 94 tournaments.

Opponent analysis distinguishes between frequently faced opponents and opponents with high win percentages. Tournament analysis similarly considers both match frequency and win rate, with a five-match threshold used for the tournament win-rate comparison.

Round analysis covers stages such as R32, R16, quarter-finals, semi-finals, and finals. Finals analysis separately identifies 67 finals, 48 wins, and 19 losses, corresponding to approximately 71.64% finals win rate.

Year and month fields allow time-based exploration, while tournament tier analysis provides another dimension for comparing performance.

A key data-quality finding was that some location and set-score fields contain missing source information. These were retained as missing rather than filled with assumptions.

10. Project Outcome & Conclusion

The project successfully transformed raw badminton match records into a complete end-to-end analytics solution: data preparation, Excel analysis, MySQL database creation, SQL analysis, Power BI/DAX visualization, validation, and publishing.

The final deliverables include a cleaned analytical dataset, the MySQL **lee_singles** table, SQL analysis covering fundamental and advanced techniques, and a three-page interactive Power BI dashboard.

The project demonstrates practical skills in Excel, MySQL, SQL, Power BI, DAX, data validation, visual design, filtering, and dashboard publishing.

Overall, the project demonstrates the ability to transform raw data into structured, validated, and interactive analytical insights while maintaining consistency across multiple analytical tools.

END OF PROJECT DOCUMENTATION

